VNR Vignana Jyothi Institute of Engineering

Electronics and Communication Engineering | Course Code: EC301

Digital Signal Processing & Filter Design - Comprehensive Syllabus Lectu

Syllabus Focus: EC301 Full Syllabus Comprehensive Lecture Notes Units 1-5

Summary: Complete syllabus lecture notes covering Digital Signal Processing & Filter Design. In

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Discrete Fourier Transform & FFT

" Syllabus Scope: DFT properties, circular convolution vs linear convolution, Decimation-In-Time DIT and

Unit 2: Infinite Impulse Response IIR Filter Design

" Syllabus Scope: Analog Butterworth and Chebyshev filter approximations, Bilinear Transformation, Impuls

EC301: Digital Signal Processing & Filter Design - Advanced Topics & Assessment

Academic Track: Electronics and Communication Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Finite Impulse Response FIR Filter Design

" Linear phase conditions, symmetric/antisymmetric impulse responses, windowing methods Rectangular, Hamming,

Unit 4: Digital Filter Realization Structures

" Direct Form I, Direct Form II, Cascade, Parallel, and Lattice ladder realizations for IIR and FIR architectu

Unit 5: Finite Word Length Effects in DSP

" Fixed vs floating-point representations, ADC quantization noise, coefficient round-off error, limit cycle os

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

1. Compute the 8-point DFT using Radix-2 DIT-FFT butterfly signal flow graph with twiddle factor computat
2. Design a digital low-pass Butterworth filter with passband edge 0.2*pi and stopband edge 0.3*pi using
3. Formulate the filter coefficients of a 7-tap linear phase FIR band-pass filter using the Hamming windo

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* John G. Proakis, Dimitris G. Manolakis - Digital Signal Processing: Principles, Algorithms, and Applica
* Alan V. Oppenheim, Ronald W. Schafer - Discrete-Time Signal Processing Pearson 3rd Ed

Authorized NoteVault Educational Document " Verified for Registered University Students